Column Dictionary

Dataset: df_complete_full_analysis_ready.csv

Study: Insulin response and protein intake are associated with distinct aspects of weight loss during obesity treatment

Timepoints: W0 = Baseline | W8 = Week 8

Section 1 — Participant Identifiers

Column	Description	Type	Values / Unit
client_id	De-identified participant identifier	Character	Unique ID
date_w0	Date of baseline assessment	Date	YYYY-MM-DD
date_w8	Date of week-8 assessment	Date	YYYY-MM-DD

Section 2 — Demographics

Column	Description	Type	Values / Unit
age	Age at baseline	Continuous	Years
gender	Biological sex	Categorical	"male" / "female"
diabetes	Type 2 diabetes (T2D) status at baseline	Categorical	"Yes" / "No"
diabetes_label	T2D label (derived from diabetes)	Categorical	"T2DM" / "non-T2DM"

Section 3 — Body Composition (Baseline and Week 8)

Measured by bioelectrical impedance analysis (InBody270; InBody Co., Ltd.) after ≥8 h overnight fast.

Column	Description	Unit
weight_w0 / weight_w8	Body weight	kg
bmi_w0 / bmi_w8	Body mass index	kg/m²
waist_circumference_w0 / _w8	Conventional waist circumference	cm
body_fat_mass_w0 / _w8	Fat mass	kg
body_fat_mass_percentage_w0 / _w8	Fat mass percentage	%
weight_without_fat_w0 / _w8	Fat-free mass	kg
muscle_mass_w0 / _w8	Skeletal muscle mass	kg
muscle_mass_percentage_w0 / _w8	Skeletal muscle mass percentage	%

Section 4 — Metabolic Markers (Baseline and Week 8)

Venous blood collected after ≥8 h overnight fast; assays performed at Le Zen Reference Laboratory.

Column	Description	Unit
hba1c_w0 / _w8	Glycated haemoglobin (HbA1c)	%
glucose_ac_w0 / _w8	Fasting plasma glucose	mg/dL
insulin_w0 / _w8	Fasting insulin	µU/mL
homa_ir_w0 / _w8	HOMA-IR	—
triglyceride_w0 / _w8	Triglycerides	mg/dL
total_cholesterol_w0 / _w8	Total cholesterol	mg/dL
high_density_lipoprotein_w0 / _w8	HDL cholesterol (HDL-C)	mg/dL
low_density_lipoprotein_cholesterol_w0 / _w8	LDL cholesterol (LDL-C)	mg/dL
systolic_blood_pressure_w0 / _w8	Systolic blood pressure	mmHg

Column	Description	Unit
diastolic_blood_pressure_w0 / _w8	Diastolic blood pressure	mmHg

HOMA-IR = Fasting insulin (μU/mL) × Fasting glucose (mg/dL) / 405

Section 5 — OGTT Measurements (Baseline Only)

Oral glucose tolerance test performed at baseline as part of routine clinical assessment.

Column	Description	Unit
glucose_pc_2hr_w0	2-hour OGTT plasma glucose at baseline	mg/dL
insulin_pc_1hr_w0	1-hour OGTT insulin at baseline	μU/mL
insulin_pc_2hr_w0	2-hour OGTT insulin at baseline	μU/mL

Section 6 — Dietary Intake (8-Week Program Average)

Estimated from dietitian-reviewed meal image records via the COFIT mobile application. Values represent participant-level mean daily intake across all dietary records during the 8-week intervention.

Column	Description	Unit
total_energy	Mean daily total energy intake	kcal/day
diet_w8_protein	Mean daily protein intake	g/day
diet_w8_fat	Mean daily fat intake	g/day
diet_w8_carbohydrate	Mean daily carbohydrate intake	g/day
protein_per_kg	Protein intake normalized to baseline body weight	g/kg/day

Section 7 — Disease Classifications

Column	Description	Timepoint
met_gp	Metabolic syndrome classification	Baseline (W0)
met_gp_w8	Metabolic syndrome classification	Week 8 (W8)
hlp_gp	Hyperlipidemia classification	Baseline (W0)
hlp_gp_w8	Hyperlipidemia classification	Week 8 (W8)

Metabolic syndrome (modified NCEP-ATP III for Asian populations): ≥3 of — waist ≥90 cm (men) / ≥80 cm (women); TG ≥150 mg/dL; HDL-C <40 (men) / <50 (women) mg/dL; SBP ≥130 or DBP ≥85 mmHg; fasting glucose ≥100 mg/dL.

Hyperlipidemia (NCEP-ATP III): any of — total cholesterol ≥200, LDL-C ≥130, TG ≥150, or HDL-C <40 (men) / <50 (women) mg/dL.

Section 8 — Derived / Calculated Variables

Column	Description	Formula	Unit
weight_loss_pct	Percentage weight change W0→W8	(W8-W0)/|W0|×100	% (negative = loss)
muscle_change_pct	Percentage muscle mass change W0→W8	(W8-W0)/|W0|×100	%
insulin_change	Absolute fasting insulin change W0→W8	W8-W0	μU/mL
insulin_change_pct	Percentage fasting insulin change W0→W8	(W8-W0)/|W0|×100	% (neg. = improvement)
carb_pct_energy	Carbohydrate as % of total energy	carb×4/energy×100	%
fat_pct_energy	Fat as % of total energy	fat×9/energy×100	%
delta_[variable]	Change from W0 to W8 for any variable	W8-W0	Same as source

Section 9 — 3D Body Circumferences (19 Sites)

Measured by TG2000-F 3D body scanner (TG3D Studio Inc.). Column names use CamelCase dot notation. Available as [BaseName]_w0, [BaseName]_w8, and delta_[BaseName] (W8-W0). All values in cm.

Base Column Name	Anatomical Site	Bilateral
Collar.Circumference	Collar	No
Neck.Base.Circumference	Neck base	No
Chest.Circumference	Chest	No
Left/Right.Upper.Arm.Circumference	Upper arm	Yes (L/R)
Left/Right.Wrist.Circumference	Wrist	Yes (L/R)
Narrow.Waist.Circumference	Narrow waist	No
Belly.Circumference	Belly (abdominal)	No
High.Hip.Circumference	High hip	No
Low.Hip.Circumference	Low hip	No
Left/Right.Thigh.Circumference	Thigh	Yes (L/R)
Left/Right.Knee.Circumference	Knee	Yes (L/R)
Left/Right.Calf.Circumference	Calf	Yes (L/R)
Left/Right.Ankle.Circumference	Ankle	Yes (L/R)

Notes on Missing Data

- *Missing values are represented as NA.*
- *Complete-case analysis was applied for each regression model independently; sample sizes per model are reported in exploratory_sample_sizes.csv.*
- *Participants missing any of: body weight, waist circumference, fasting glucose, fasting insulin, HbA1c, triglycerides, total cholesterol, HDL-C, or LDL-C were excluded during data preparation.*
- *~2% of participants had ≥1 missing 3D circumference value due to scanner detection failures and were excluded from UMAP visualization (Methods Section 4.5.4).*